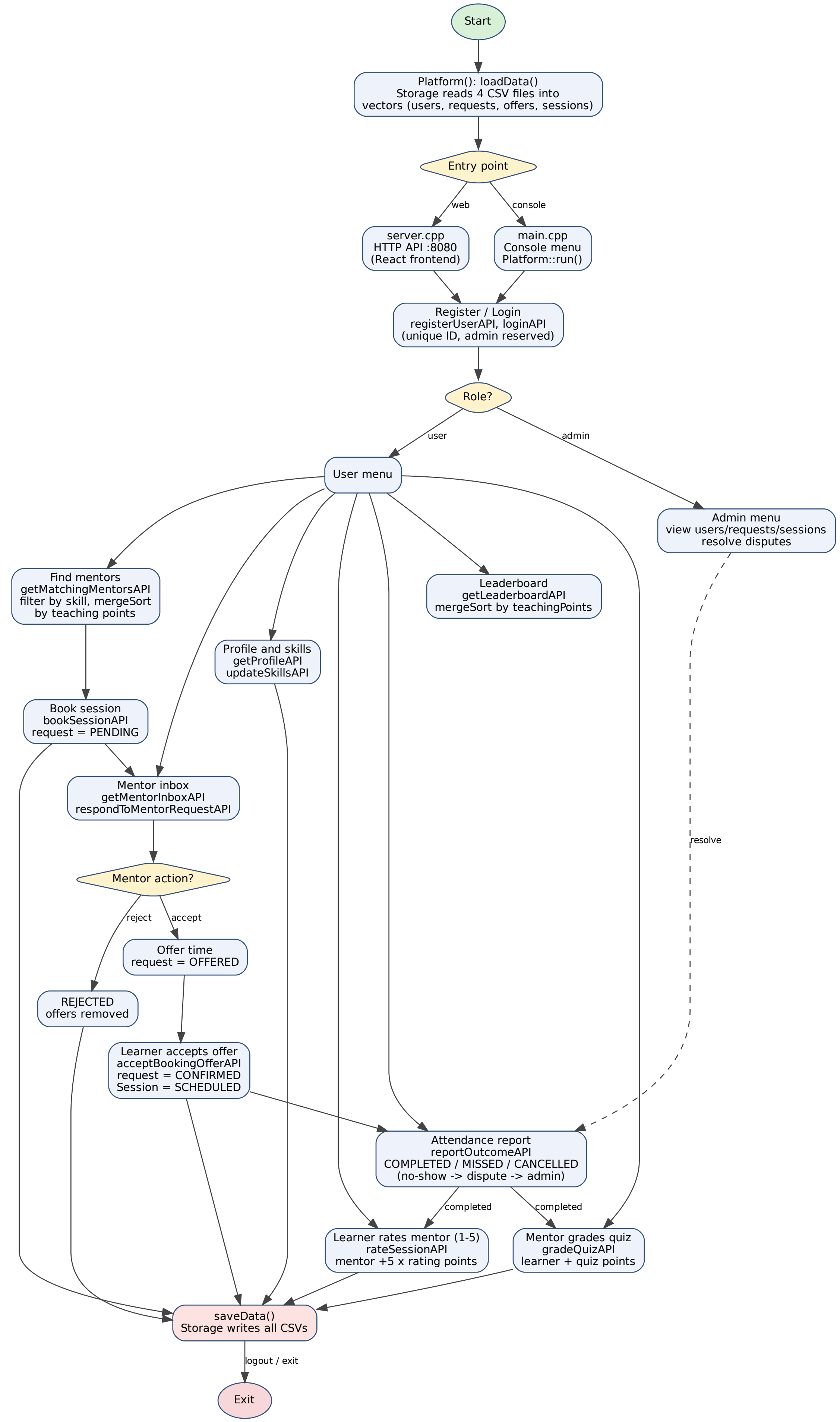
Start
Platform(): loadData()
Storage reads 4 CSV files into
vectors (users, requests, offers, sessions)
Entry point
web
console
server.cpp
HTTP API :8080
(React frontend)
main.cpp
Console menu
Platform::run()
Register / Login
registerUserAPI, loginAPI
(unique ID, admin reserved)
Role?
user
admin
User menu
Admin menu
view users/requests/sessions
resolve disputes
Find mentors
getMatchingMentorsAPI
filter by skill, mergeSort
by teaching points
Profile and skills
getProfileAPI
updateSkillsAPI
Leaderboard
getLeaderboardAPI
mergeSort by teachingPoints
Book session
bookSessionAPI
request = PENDING
Mentor inbox
getMentorInboxAPI
respondToMentorRequestAPI
Mentor action?
reject
accept
Offer time
request = OFFERED
REJECTED
offers removed
Learner accepts offer
acceptBookingOfferAPI
request = CONFIRMED
Session = SCHEDULED
resolve
Attendance report
reportOutcomeAPI
COMPLETED / MISSED / CANCELLED
(no-show -> dispute -> admin)
completed
completed
Learner rates mentor (1-5)
rateSessionAPI
mentor +5 x rating points
Mentor grades quiz
gradeQuizAPI
learner + quiz points
saveData()
Storage writes all CSVs
logout / exit
Exit